Outcrop overview



Outcrop overview



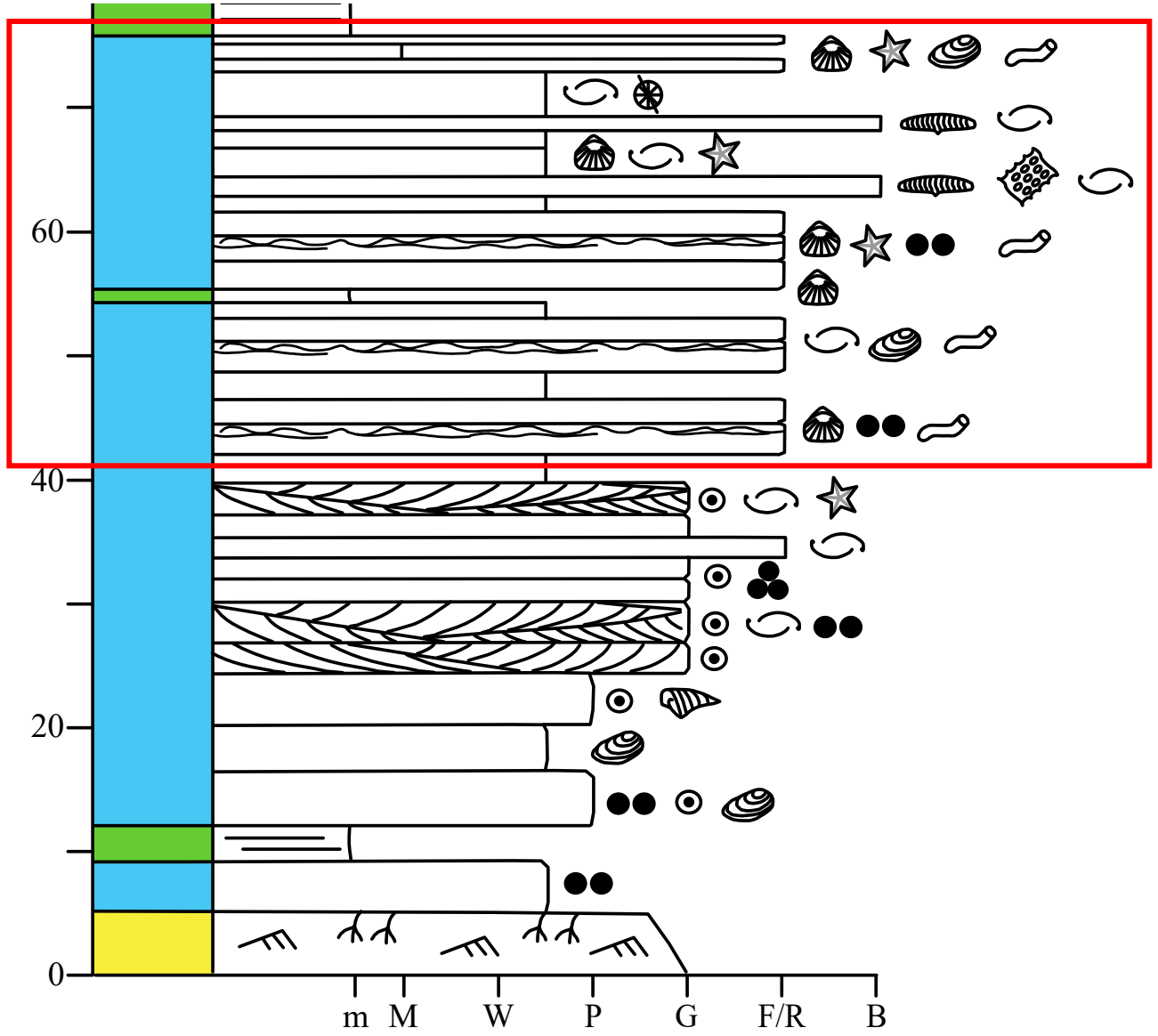
Unit 2: Somalirhynchia member

Study the following data set.

- Outcrop photos
- Thin sections
- There are also some body fossils collected from this unit for you to determine.

Summarize your observations on this unit:

- What grains were identified?
- What sedimentary structures were identified?
- What is the texture of these beds?
- What sedimentary processes can you interpret?
- What environmental information is contained within the skeletal grains?
- Overall, what environment is likely?



What is the bedding like?
Why do tops of beds appear mottled? bioturbation



(2.1)



What is the bedding like?
Why do tops of beds appear mottled? bioturbation





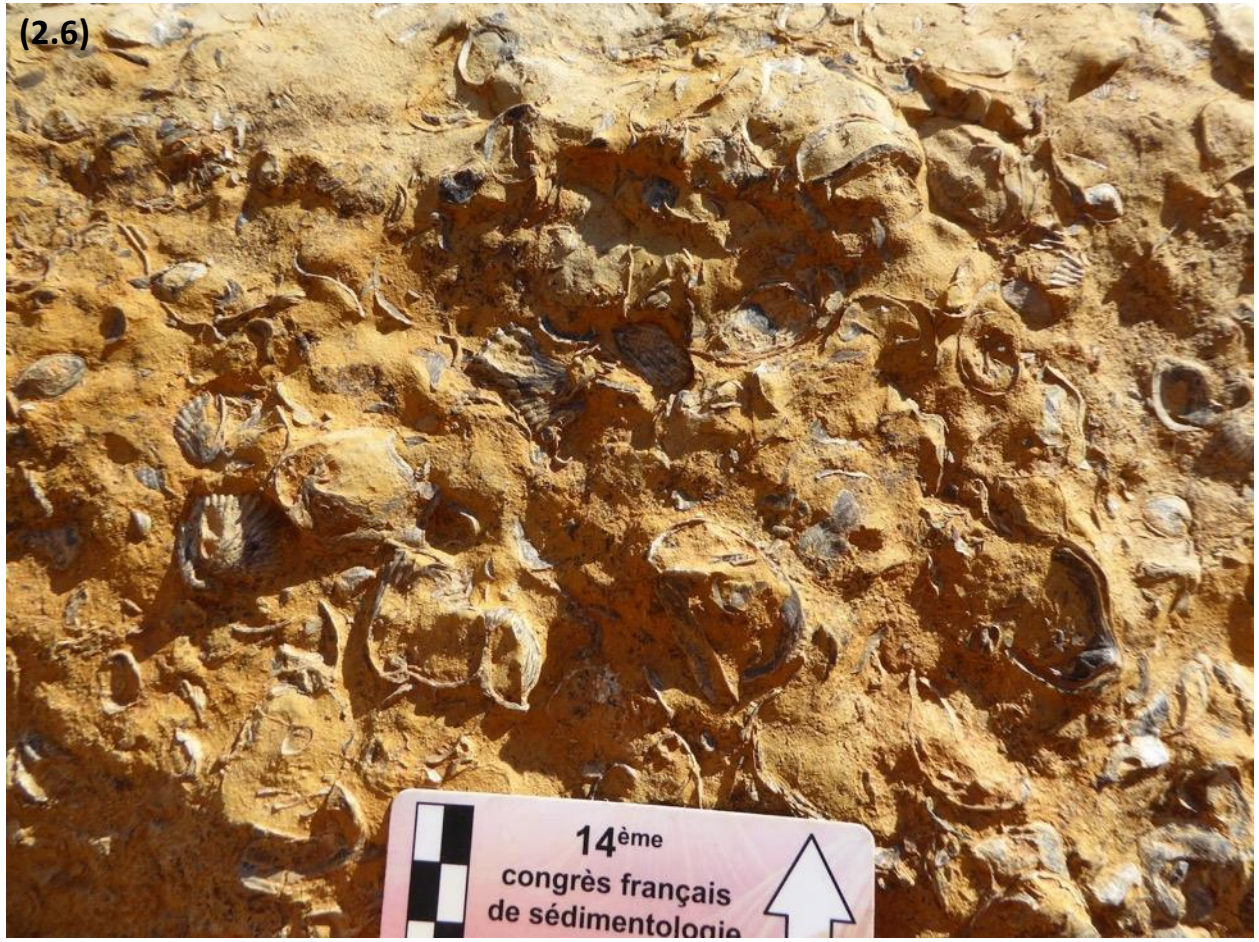
What is the bedding like?

Why do tops of beds appear mottled?

bioturbation



(2.6)

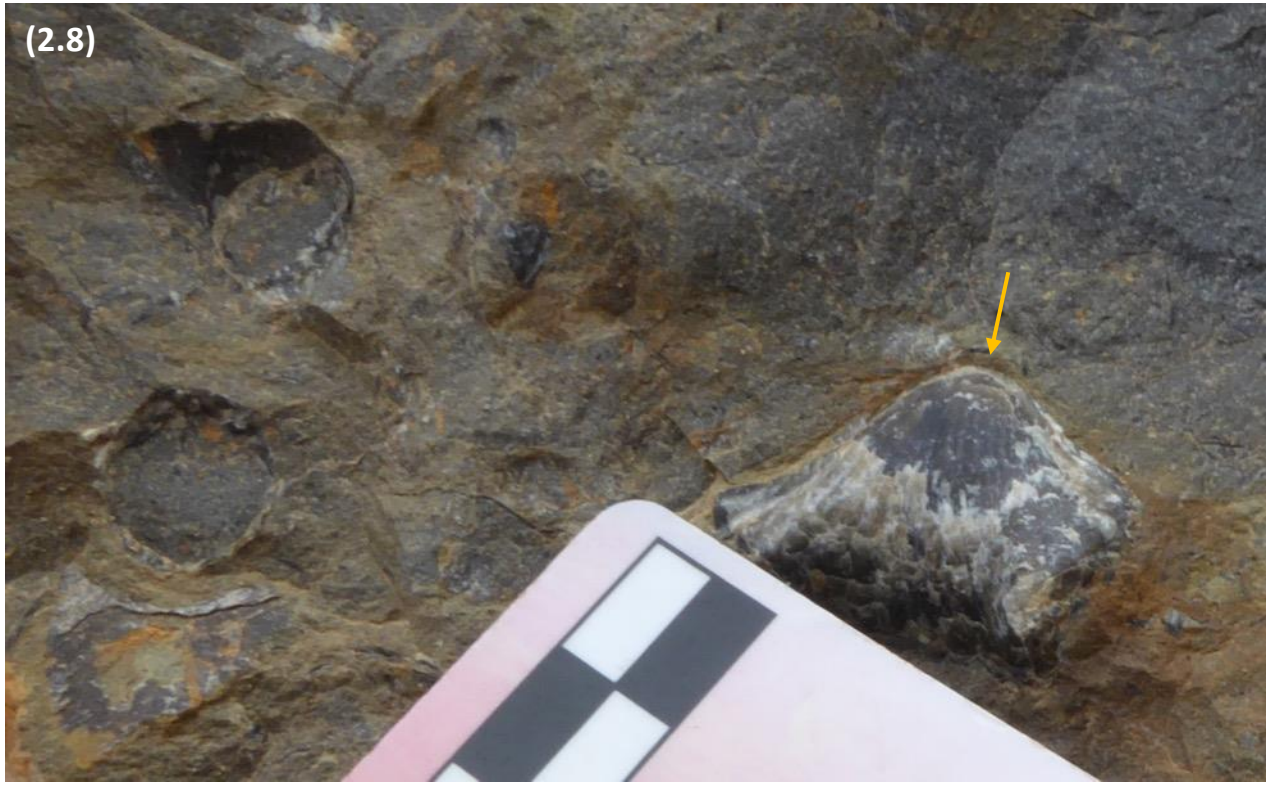


This is the main grain in these beds. What is it? Note the shapes.

brachiopods

(2.7)







This is the main grain in these beds. What is it? Note the shapes.

brachiopods

Second important grain
- identification? Echinoderms (whole)







Second important grain
- identification?
Echinoderms (whole) and spine



(2.14)



Solitary coral

Nerinea gastropods

(2.15)



(2.16)



Bivalves

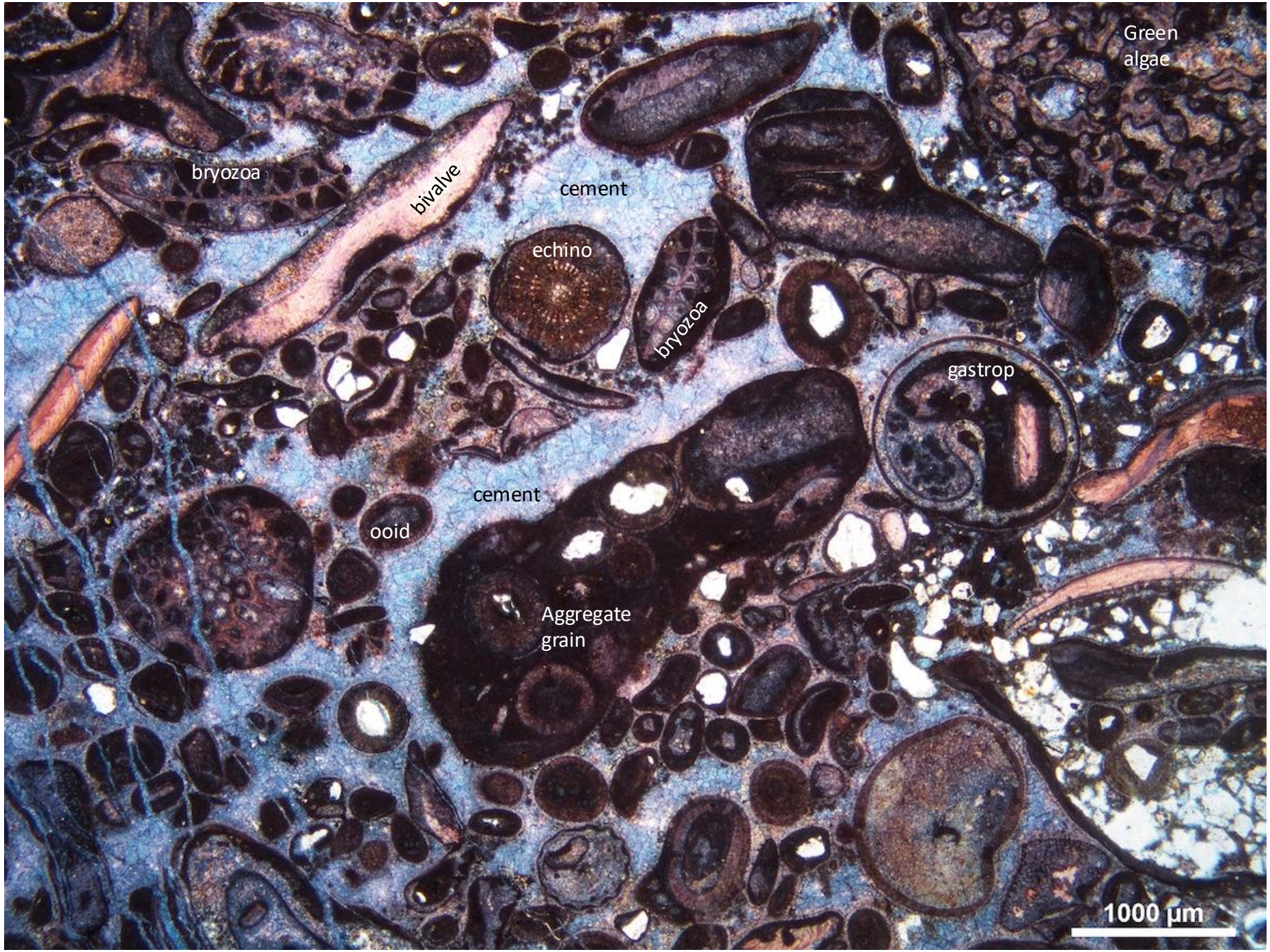
Additional grains - identification?

Complement your outcrop observations with the following thin sections:
L272 – grainstone with shell fragments (bivalves, brachiopods), bryozoan, ooids, peloids; deposited in shallow, energetic, light-rich water
L337
L340

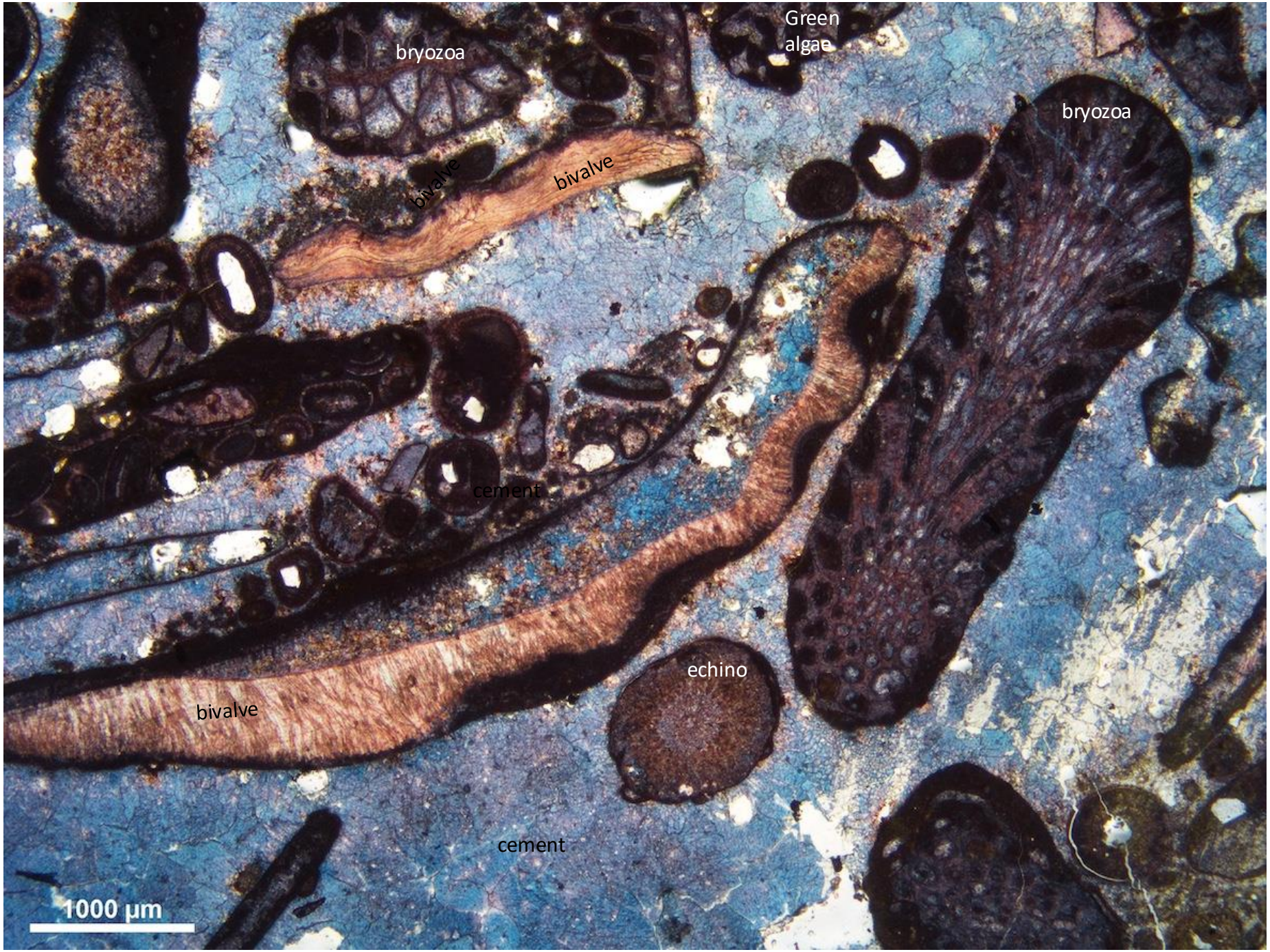
You can also study if time allows:
AS21
AS26

Take note of:
- Grains, their %, size and degree of preservation
- Presence of sparite, micrite and/or porosity, and their %
- Texture
- What is the Dunham name for this thin section?
- Make sketches of relevant features

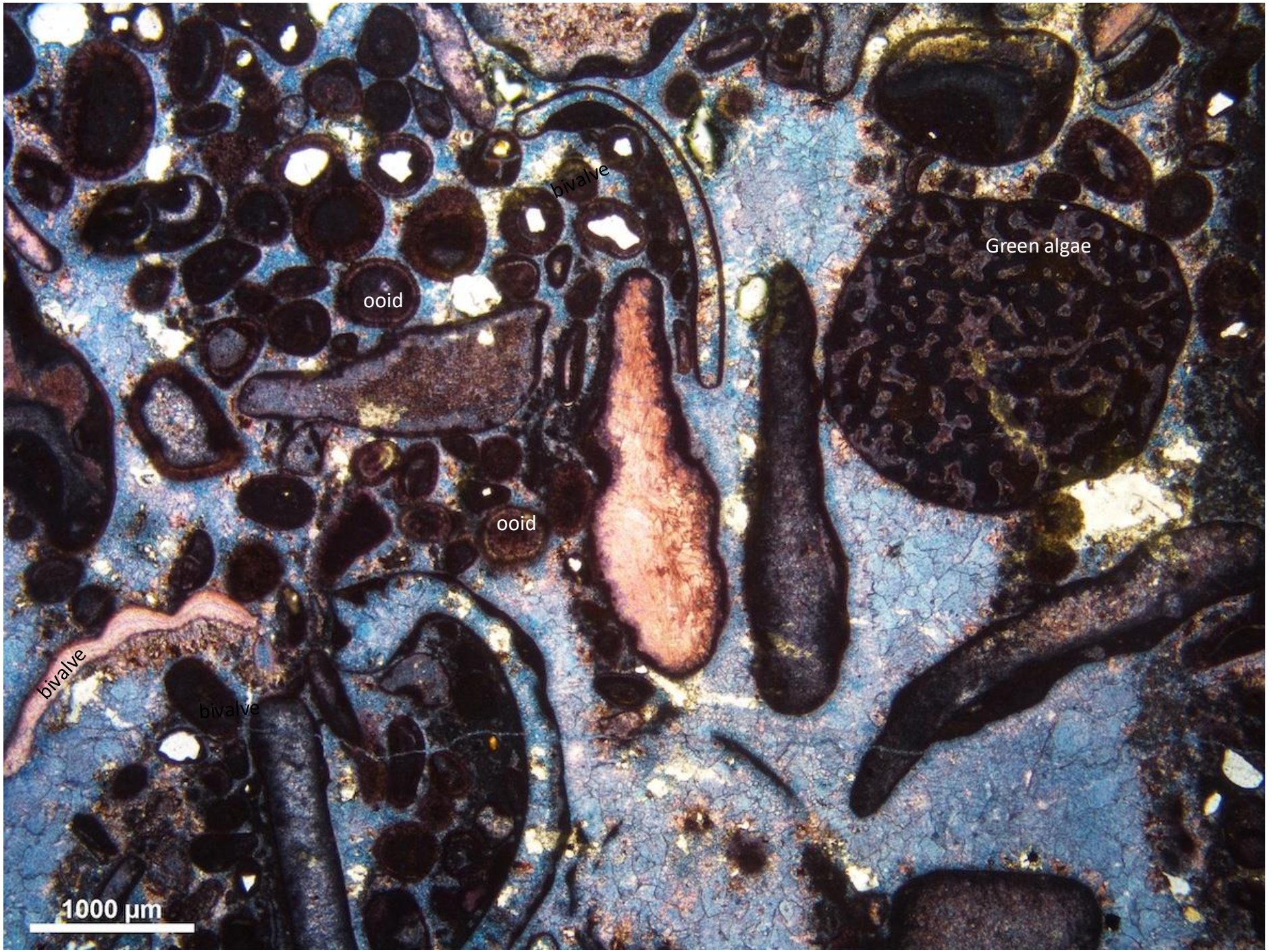
L272


Green algae
bryozoa
bivalve
cement
echino
bryozoa
gastrop
cement
ooid
Aggregate grain
1000 µm

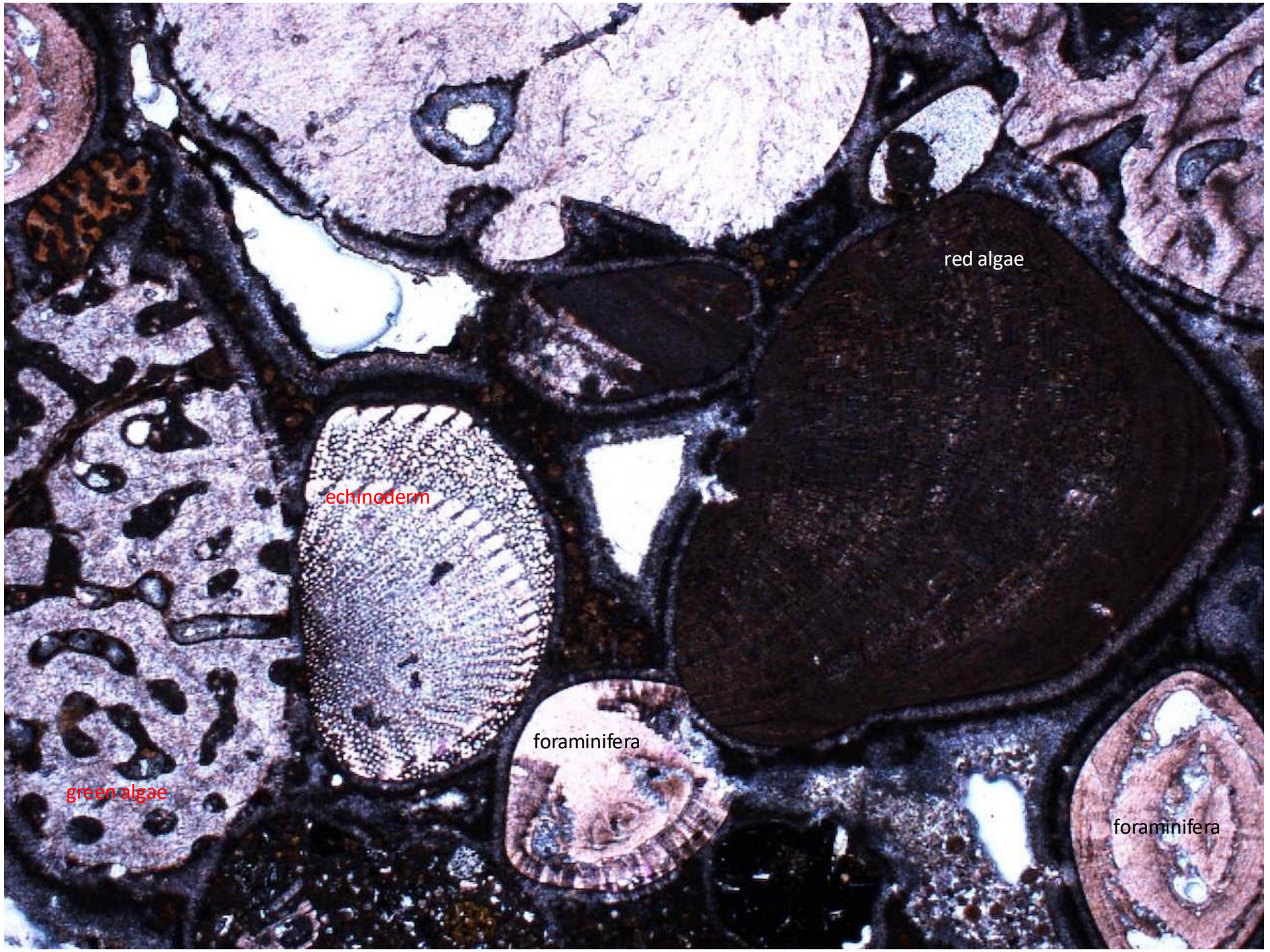
L272


bryozoa
Green algae
bryozoa
bivalve
bivalve
cement
bivalve
echino
cement
1000 µm

L272



L337


red algae
echinoderm
foraminifera
green algae
foraminifera

L340
Nicely shows the difference between peloids made from micrite and sparite; note peloids are often fused in this sample, giving the impression of just micrite matrix

